Solar System Models · Tier Pack · Aspire

Learning goal: I can use solar system data and a scale model to link distance from the Sun to orbit time.

Aspire Route (Book First)

Explain why distance and planet size cannot share the same scale on a 1 m strip, and note how a planet's atmosphere (like Venus's thick carbon dioxide clouds) can affect surface temperature.

Help: On a 1 m scale where 4,500 million km = 100 cm, Earth's 12,742 km diameter would be less than __ mm. Also, while orbit time follows distance strictly, temperature can be affected by __.

Optional Make It · Aspire

Make a Dual-Scale Critique Card

Show why planet sizes and orbit distances cannot use the same 1 m scale, and explain why Venus (second planet) is hotter than Mercury despite being further from the Sun.

Word Help

orbit time: the time a planet takes to travel once all the way around the Sun (its year)

scale model: a model that shrinks huge real distances or sizes using a fixed rule

astronomical data: measurements of planets such as distance from the Sun, diameter, and orbit time

limitation: something a science model leaves out or

cannot show accurately

Exit Check

Which planet takes longer to orbit the Sun: Earth (365 days) or Neptune (165 years)?

State the pattern linking a planet's distance from the Sun to its orbit time.

Give one strength and one limitation of our 100 cm paper strip model.